SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: MLA03 **COURSE CODE: Reinforcement learning for Environment and Agent**

COURSE OUTCOME COVERED

CO1: *Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)*

Assessment Tool Weightage: 30%

Dr Dumpala Prasanth

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Case Study **Duration: 2hrs**

1. A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. **(5 Marks)**

2. A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. **(5 Marks)**

3. An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. **(5 Marks)**

4. A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. **(5 Marks)**

5. A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. **(5 Marks)**

6. A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. **(5 Marks)**

7. A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance.**(5 Marks)**

8. A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. **(5 Marks)**

9. A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. **(5 Marks)**

10. A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. **(5 Marks)**

Code of Conduct:

I B V Chaitanya(192425376) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: chaitanya

RUBRICS FOR CASESTUDY

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

CO-1 AT-2 Case Study

Question 1: RL-Based Warehouse Robot

Understanding of the Case

A logistics company aims to develop an intelligent warehouse robot that can pick, pack, and transport items efficiently. The robot must avoid obstacles, minimize delays, and optimize delivery time. Reinforcement Learning (RL) enables the robot to learn the best actions through trial and error by interacting with the warehouse environment.

Application of Reinforcement Learning Concepts

Markov Decision Process (MDP)

States (S):

- Robot position
- Item location
- Destination location
- Battery level
- Obstacle positions
- Load status (carrying item or not)

Actions (A):

- Move forward
- Move backward
- Turn left/right
- Pick item
- Drop item
- Recharge battery
- Wait

Rewards (R):

- +30 for successful delivery

- +10 for correct item pickup
- +5 for shortest path
- -20 for collision
- -10 for wrong pickup
- -5 for unnecessary movement
- -25 for battery depletion

Policy

The robot selects actions that maximize long-term rewards while ensuring safety and efficiency.

Environment

The warehouse consists of shelves, packages, workers, charging stations, and dynamic obstacles.

Analysis

The reward function significantly affects the robot's behavior. Positive rewards encourage efficient deliveries, while penalties reduce unsafe actions. If speed is rewarded more than safety, the robot may collide with obstacles. A balanced reward function promotes safe navigation, accurate item handling, and efficient task completion.

Solution Design

A **Deep Q-Network (DQN)** is suitable because it handles large and dynamic state spaces. The robot learns through an ε-greedy strategy, balancing exploration of new paths with exploitation of the best-known routes.

Conclusion

Modeling the warehouse robot as an MDP with a balanced reward function enables efficient, safe, and accurate warehouse operations, improving productivity and reducing operational costs.

Question 2: Movie Recommendation System

Understanding of the Case

A streaming platform wants to recommend movies dynamically based on user preferences and viewing history. The goal is to maximize user engagement, watch time, and customer satisfaction using Reinforcement Learning.

Application of Reinforcement Learning Concepts

State (S)

- User profile
- Viewing history
- Preferred genres
- Ratings
- Watch duration
- Time of day

Actions (A)

- Recommend a movie
- Recommend a TV series
- Recommend trending content
- Recommend a new genre

Reward (R)

- +15 for watching a recommended movie completely
- +10 for a positive rating
- +5 for adding to the watchlist
- -10 if the recommendation is skipped
- -15 if the user exits the platform

Policy (π)

The recommendation system learns to select content that maximizes long-term user satisfaction and engagement.

Environment

Users interact with the streaming platform by watching, rating, skipping, or searching for content.

Analysis

Reward signals influence recommendation quality. Positive rewards increase recommendations similar to user interests, while negative rewards discourage irrelevant suggestions. Exploration introduces new content, whereas exploitation recommends familiar favorites. A balance between both prevents repetitive recommendations and improves content discovery.

Solution Design

A **Contextual Bandit** or **Deep Q-Network (DQN)** algorithm can efficiently learn personalized recommendations based on user interactions.

Conclusion

Reinforcement Learning enables adaptive and personalized movie recommendations, increasing user engagement and overall platform performance.

Question 3: Smart Agriculture System

Understanding of the Case

An autonomous agricultural system uses Reinforcement Learning to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. The objective is to improve crop yield while conserving water and fertilizer.

Application of Reinforcement Learning Concepts

States (S)

- Soil moisture
- Soil nutrients
- Temperature
- Humidity
- Weather forecast
- Crop growth stage

Actions (A)

- Increase irrigation

- Reduce irrigation
- Apply fertilizer
- Delay irrigation
- Maintain current settings

Rewards (R)

- +30 for healthy crop growth
- +15 for water conservation
- +10 for efficient fertilizer usage
- -20 for crop damage
- -15 for over-irrigation
- -10 for fertilizer wastage

Policy (π)

The system selects irrigation and fertilizer levels that maximize crop productivity while minimizing resource consumption.

Environment

The environment includes farmland, changing weather conditions, soil characteristics, and crop growth.

Analysis

Crop yield is a delayed reward because results become visible only after several weeks. Temporal Difference learning helps the system estimate future rewards before final outcomes are observed. Efficient exploration enables better adaptation to seasonal weather changes.

Solution Design

A **SARSA** or **Deep Q-Network (DQN)** algorithm is appropriate because it can learn optimal agricultural decisions from continuous environmental feedback.

Conclusion

An RL-based agricultural system improves crop production, conserves natural resources, and supports sustainable farming practices.

Question 4: Cybersecurity Intrusion Detection

Understanding of the Case

A cybersecurity system uses Reinforcement Learning to detect and respond to cyberattacks in real time. The objective is to identify malicious activities while minimizing false alarms.

Application of Reinforcement Learning Concepts

States (S)

- Network traffic
- Login attempts
- CPU usage
- IP address reputation
- Packet size
- Connection status

Actions (A)

- Allow traffic
- Block IP address
- Isolate infected device
- Alert security administrator
- Monitor suspicious activity

Rewards (R)

- +30 for correctly detecting an attack
- +15 for preventing intrusion
- -25 for false alarms
- -35 for missed attacks
- -10 for unnecessary blocking

Policy (π)

The system chooses actions that maximize network security while reducing operational disruptions.

Environment

The environment includes network devices, servers, users, applications, and continuously evolving cyber threats.

Analysis

Cyberattacks occur infrequently, resulting in sparse rewards. Real-time decision-making requires the system to process large volumes of network traffic with low latency. Poor reward design may increase false positives or allow attacks to go undetected.

Solution Design

A **Deep Q-Network (DQN)** is suitable because it efficiently handles complex network states and continuously adapts to new attack patterns.

Conclusion

A properly designed RL-based intrusion detection system enhances cybersecurity by providing accurate, adaptive, and real-time threat detection.

Question 5: Ride-Sharing Driver Allocation

Understanding of the Case

A ride-sharing company aims to optimize driver allocation and dynamic pricing using Reinforcement Learning. The system should minimize passenger waiting time, maximize driver utilization, and maintain fair pricing.

Application of Reinforcement Learning Concepts

States (S)

- Driver locations
- Customer requests
- Traffic conditions
- Weather conditions
- Time of day
- Current pricing level

Actions (A)

- Assign a driver
- Reposition idle drivers
- Increase fare
- Reduce fare
- Delay assignment

Rewards (R)

- +30 for successful ride completion
- +20 for reduced waiting time
- +10 for efficient driver utilization
- -15 for ride cancellation
- -20 for long customer waiting time
- -25 for customer complaints

Policy (π)

The system learns the best driver assignment and pricing strategy that maximizes long-term revenue while maintaining customer satisfaction.

Environment

The environment consists of passengers, drivers, road traffic, weather conditions, and fluctuating customer demand.

Analysis

Environmental uncertainty caused by traffic congestion, weather changes, and demand fluctuations affects learning performance. Dynamic pricing improves efficiency but may create ethical concerns if prices become excessively high during emergencies. Fair pricing policies are necessary to maintain customer trust.

Solution Design

A **Deep Q-Network (DQN)** combined with demand prediction techniques can optimize both driver allocation and pricing decisions while adapting to changing conditions.

Question 6: Smart Home Automation System

Understanding of the Case

A smart home automation system uses Reinforcement Learning (RL) to control lighting, heating, and cooling based on occupancy and weather conditions. The objective is to improve user comfort while reducing energy consumption and electricity costs.

Application of Reinforcement Learning Concepts

States (S)

- Room occupancy
- Indoor temperature
- Outdoor weather
- Time of day
- Humidity
- Current energy consumption

Actions (A)

- Turn lights ON/OFF
- Increase or decrease air conditioner temperature
- Turn heater ON/OFF
- Open or close smart windows
- Activate energy-saving mode

Rewards (R)

- +20 for maintaining user comfort
- +15 for reducing energy consumption
- +10 for using renewable energy efficiently
- -20 for user discomfort
- -15 for unnecessary electricity usage
- -10 for frequent device switching

Policy (π)

The system learns to choose actions that maximize user comfort while minimizing energy usage.

Environment

The environment includes home appliances, occupants, weather conditions, and electricity supply.

Analysis

Comfort and energy efficiency are conflicting objectives. A balanced reward function ensures both objectives are achieved. Reward shaping encourages the agent to learn desirable behaviors faster by assigning intermediate rewards for energy-efficient actions.

Solution Design

A Deep Q-Network (DQN) or SARSA algorithm can continuously learn user preferences and adapt appliance control strategies.

Conclusion

An RL-based smart home improves comfort, reduces energy consumption, lowers electricity costs, and supports sustainable living through intelligent automation.

Question 7: RL-Based Strategy Game Agent

Understanding of the Case

A gaming company is developing an RL agent capable of playing a complex strategy game against human players. The agent must learn long-term strategies rather than relying on immediate rewards.

Application of Reinforcement Learning Concepts

States (S)

- Current game board
- Player resources
- Unit positions
- Enemy strength
- Health points
- Remaining game time

Actions (A)

- Attack enemy
- Defend position

- Gather resources
- Build units
- Upgrade technology
- Retreat strategically

Rewards (R)

- +100 for winning the game
- +30 for capturing important objectives
- +20 for defeating enemy units
- -20 for losing important units
- -100 for losing the game

Policy (π)

The policy learns to maximize long-term rewards by selecting actions that improve the probability of winning.

Environment

The environment consists of the game map, opponents, available resources, and changing game conditions.

Analysis

Long-term rewards encourage strategic planning rather than short-term gains. Exploration strategies such as ε-greedy allow the agent to discover stronger strategies, while exploitation improves performance using learned knowledge.

Solution Design

A Deep Q-Network (DQN) or Policy Gradient algorithm is suitable because the game has a large state and action space requiring advanced decision-making.

Conclusion

Reinforcement Learning enables the game agent to learn intelligent strategies, improve adaptability, and compete effectively against human players.

Question 8: Fraud Detection System

Understanding of the Case

A financial institution uses Reinforcement Learning to detect fraudulent transactions in real time. The objective is to maximize fraud detection while minimizing false alarms and financial losses.

Application of Reinforcement Learning Concepts

States (S)

- Transaction amount
- Customer transaction history
- Location
- Device information
- Transaction frequency
- Risk score

Actions (A)

- Approve transaction
- Reject transaction
- Flag for manual review
- Request additional authentication

Rewards (R)

- +30 for correctly detecting fraud
- +15 for correctly approving genuine transactions
- -30 for false rejection
- -40 for missed fraudulent transactions
- -20 for unnecessary manual reviews

Policy (π)

The policy selects actions that maximize fraud detection accuracy while minimizing customer inconvenience.

Environment

The environment includes customers, banking systems, transaction networks, and continuously evolving fraud techniques.

Analysis

Fraud cases are rare, resulting in highly imbalanced data. Delayed feedback occurs because fraud may only be confirmed after investigation. Incorrect decisions can cause financial losses or inconvenience legitimate customers, making reward design critical.

Solution Design

A Deep Q-Network (DQN) combined with fraud detection models enables continuous learning from new fraud patterns and improves detection accuracy.

Conclusion

An RL-based fraud detection system enhances financial security by adapting to evolving fraud methods while maintaining a balance between security and customer satisfaction.

Question 9: Personalized Rehabilitation System

Understanding of the Case

A healthcare system uses Reinforcement Learning to recommend personalized rehabilitation exercises based on each patient's recovery progress. The objective is to improve recovery while ensuring patient safety and comfort.

Application of Reinforcement Learning Concepts

States (S)

- Patient mobility
- Pain level
- Exercise history
- Recovery progress
- Age
- Medical condition

Actions (A)

- Increase exercise intensity
- Reduce intensity
- Recommend a new exercise

- Continue current exercise
- Recommend rest

Rewards (R)

- +30 for improvement in mobility
- +20 for successful exercise completion
- +10 for patient satisfaction
- -30 for injury
- -20 for excessive pain
- -15 for poor recovery progress

Policy (π)

The system learns personalized rehabilitation plans that maximize recovery while minimizing discomfort and injury.

Environment

The environment consists of patients, healthcare professionals, rehabilitation equipment, and medical conditions.

Analysis

Recovery improvements are delayed, making long-term rewards important. Patient variability requires personalized learning since different patients respond differently to the same exercise. Ethical concerns include patient safety, privacy, fairness, and ensuring that medical professionals supervise treatment decisions.

Solution Design

A SARSA or Deep Q-Network (DQN) algorithm with clinician supervision is appropriate because it continuously adapts treatment recommendations while ensuring patient safety.

Conclusion

Reinforcement Learning enables personalized rehabilitation programs that improve recovery outcomes while maintaining ethical standards and patient safety.

Question 10: Smart Grid Electricity Distribution

Understanding of the Case

A smart grid system uses Reinforcement Learning to manage electricity distribution, reduce peak load demand, and improve energy efficiency. The objective is to provide reliable electricity while minimizing energy wastage.

Application of Reinforcement Learning Concepts

Markov Decision Process (MDP)

States (S)

- Current electricity demand
- Grid load
- Renewable energy availability
- Battery storage level
- Electricity price
- Weather conditions

Actions (A)

- Increase power generation
- Reduce power supply
- Store energy
- Release stored energy
- Shift non-critical loads
- Purchase external electricity

Rewards (R)

- +30 for balanced electricity distribution
- +20 for reducing peak demand
- +15 for efficient renewable energy utilization
- -30 for power outages
- -20 for energy wastage
- -15 for grid instability

Policy (π)

The policy learns the optimal electricity distribution strategy that maximizes efficiency and reliability.

Environment

The environment includes consumers, power plants, renewable energy sources, batteries, transmission lines, and weather conditions.

Analysis

Large-scale power networks create scalability challenges due to millions of devices and continuous data generation. Real-time learning is essential because electricity demand changes rapidly throughout the day. Efficient reward design encourages stable power supply while reducing operational costs and improving renewable energy utilization.

Solution Design

A Deep Reinforcement Learning (DQN) algorithm is suitable because it can process large state spaces and adapt quickly to changing grid conditions. Combining RL with demand forecasting further improves system performance.

Conclusion

An RL-based smart grid provides efficient, reliable, and sustainable electricity management by balancing demand, reducing peak loads, and maximizing renewable energy usage while ensuring stable power distribution.

Reinforcement Learning helps ride-sharing companies improve operational efficiency, reduce passenger waiting time, optimize driver utilization, and maintain fair and ethical pricing strategies.